

## Subnetting Project

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, LI - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
* - replicated route, % - next hop override, p - overrides from FR
Gateway of last resort is 12.255.0.2 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 12.255.0.2
C 12.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C 12.0.0.0/24 is directly connected, Ethernet0/0
C 12.0.0.1/32 is directly connected, Ethernet0/0
C 12.255.0.0/30 is directly connected, Ethernet0/1
C 12.255.0.1/32 is directly connected, Ethernet0/1
Sep 11 07:35:52.657: NFWP-6-PNP DISCOVERY_STOPPED: PnP Disc
R2#
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, LI - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
* - replicated route, % - next hop override, p - overrides from FR
Gateway of last resort is 12.255.0.4 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 12.255.0.4
C 12.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C 12.0.0.0/24 is directly connected, Ethernet0/0
C 12.0.1.1/32 is directly connected, Ethernet0/0
C 12.255.0.4/30 is directly connected, Ethernet0/1
C 12.255.0.1/32 is directly connected, Ethernet0/1
R3#
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, LI - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
* - replicated route, % - next hop override, p - overrides from FR
Gateway of last resort is 12.255.0.10 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 12.255.0.10
C 12.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C 12.0.2.0/24 is directly connected, Ethernet0/0
C 12.0.2.1/32 is directly connected, Ethernet0/0
C 12.255.0.6/30 is directly connected, Ethernet0/1
C 12.255.0.9/32 is directly connected, Ethernet0/1
R4#
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, LI - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
* - replicated route, % - next hop override, p - overrides from FR
Gateway of last resort is not set
S 12.0.0.0/8 is variably subnetted, 19 subnets, 5 masks
S 12.0.0.0/25 [1/0] via 12.255.0.1
S 12.0.1.0/25 [1/0] via 12.255.0.5
C 12.255.0.0/30 is directly connected, Ethernet0/0
C 12.255.0.2/32 is directly connected, Ethernet0/0
C 12.255.0.4/30 is directly connected, Ethernet0/1
C 12.255.0.6/32 is directly connected, Ethernet0/1
C 12.255.0.8/30 is directly connected, Ethernet0/2
C 12.255.0.10/32 is directly connected, Ethernet0/2
C 12.255.0.12/30 is directly connected, Ethernet0/3
C 12.255.0.13/32 is directly connected, Ethernet0/3
C 12.255.0.16/30 is directly connected, Ethernet0/0
C 12.255.0.17/32 is directly connected, Ethernet0/0
S 129.70.0.0/16 is variably subnetted, 2 subnets, 2 masks
S 129.70.0.0/24 [1/0] via 12.255.0.14
S 129.70.1.0/24 [1/0] via 12.255.0.18
R5#
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, LI - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
* - replicated route, % - next hop override, p - overrides from FR
Gateway of last resort is 12.255.0.13 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 12.255.0.13
C 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 12.255.0.12/30 is directly connected, Ethernet0/0
C 12.255.0.14/32 is directly connected, Ethernet0/0
C 129.70.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 129.70.0.0/24 is directly connected, Ethernet0/1
C 129.70.0.1/32 is directly connected, Ethernet0/1
R6#
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, LI - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
* - replicated route, % - next hop override, p - overrides from FR
Gateway of last resort is 12.255.0.17 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 12.255.0.17
C 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 12.255.0.16/30 is directly connected, Ethernet0/0
C 12.255.0.18/32 is directly connected, Ethernet0/0
C 129.70.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 129.70.1.0/24 is directly connected, Ethernet0/1
C 129.70.1.1/32 is directly connected, Ethernet0/1

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QEMU (Win-1)

C:\WINDOWS\system32\cmd.exe

```

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\User>ping 12.0.1.2

Pinging 12.0.1.2 with 32 bytes of data:

Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=1ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 12.0.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\User>ping 12.0.2.2

Pinging 12.0.2.2 with 32 bytes of data:

Reply from 12.0.2.2: bytes=32 time=2ms TTL=125
Reply from 12.0.2.2: bytes=32 time=1ms TTL=125
Reply from 12.0.2.2: bytes=32 time=2ms TTL=125
Reply from 12.0.2.2: bytes=32 time=3ms TTL=125

Ping statistics for 12.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 129.70.0.2

Pinging 129.70.0.2 with 32 bytes of data:

Reply from 129.70.0.2: bytes=32 time=3ms TTL=125
Reply from 129.70.0.2: bytes=32 time=3ms TTL=125
Reply from 129.70.0.2: bytes=32 time=3ms TTL=125
Reply from 129.70.0.2: bytes=32 time=2ms TTL=125

Ping statistics for 129.70.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 129.70.1.2

Pinging 129.70.1.2 with 32 bytes of data:

Reply from 129.70.1.2: bytes=32 time=2ms TTL=125
Reply from 129.70.1.2: bytes=32 time=2ms TTL=125
Reply from 129.70.1.2: bytes=32 time=2ms TTL=125
Reply from 129.70.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 129.70.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\User>

```

C:\WINDOWS\system...

```
QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 12.0.0.2

Pinging 12.0.0.2 with 32 bytes of data:

Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=1ms TTL=125

Ping statistics for 12.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 12.0.2.2

Pinging 12.0.2.2 with 32 bytes of data:

Reply from 12.0.2.2: bytes=32 time=3ms TTL=125
Reply from 12.0.2.2: bytes=32 time=3ms TTL=125
Reply from 12.0.2.2: bytes=32 time=3ms TTL=125
Reply from 12.0.2.2: bytes=32 time=2ms TTL=125

Ping statistics for 12.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 129.70.0.2

Pinging 129.70.0.2 with 32 bytes of data:

Reply from 129.70.0.2: bytes=32 time=2ms TTL=125
Reply from 129.70.0.2: bytes=32 time=2ms TTL=125
Reply from 129.70.0.2: bytes=32 time=2ms TTL=125
Reply from 129.70.0.2: bytes=32 time=2ms TTL=125

Ping statistics for 129.70.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 129.70.1.2

Pinging 129.70.1.2 with 32 bytes of data:

Reply from 129.70.1.2: bytes=32 time=7ms TTL=125
Reply from 129.70.1.2: bytes=32 time=2ms TTL=125
Reply from 129.70.1.2: bytes=32 time=2ms TTL=125
Reply from 129.70.1.2: bytes=32 time=1ms TTL=125

Ping statistics for 129.70.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 3ms

C:\Documents and Settings\user>
```

```
QEMU (Win-3)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 12.0.0.2

Pinging 12.0.0.2 with 32 bytes of data:

Reply from 12.0.0.2: bytes=32 time=1ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=3ms TTL=125

Ping statistics for 12.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 12.0.1.2

Pinging 12.0.1.2 with 32 bytes of data:

Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=3ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=1ms TTL=125

Ping statistics for 12.0.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 129.70.0.2

Pinging 129.70.0.2 with 32 bytes of data:

Reply from 129.70.0.2: bytes=32 time=1ms TTL=125
Reply from 129.70.0.2: bytes=32 time=2ms TTL=125
Reply from 129.70.0.2: bytes=32 time=1ms TTL=125
Reply from 129.70.0.2: bytes=32 time=2ms TTL=125

Ping statistics for 129.70.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 129.70.1.2

Pinging 129.70.1.2 with 32 bytes of data:

Reply from 129.70.1.2: bytes=32 time=3ms TTL=125
Reply from 129.70.1.2: bytes=32 time=1ms TTL=125
Reply from 129.70.1.2: bytes=32 time=1ms TTL=125
Reply from 129.70.1.2: bytes=32 time=1ms TTL=125

Ping statistics for 129.70.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Documents and Settings\user>
```

```
QEMU (Win-4)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 12.0.0.2

Pinging 12.0.0.2 with 32 bytes of data:

Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=3ms TTL=125

Ping statistics for 12.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 12.0.1.2

Pinging 12.0.1.2 with 32 bytes of data:

Reply from 12.0.1.2: bytes=32 time=3ms TTL=125
Reply from 12.0.1.2: bytes=32 time=3ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=3ms TTL=125

Ping statistics for 12.0.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 12.0.2.2

Pinging 12.0.2.2 with 32 bytes of data:

Reply from 12.0.2.2: bytes=32 time=2ms TTL=125
Reply from 12.0.2.2: bytes=32 time=2ms TTL=125
Reply from 12.0.2.2: bytes=32 time=1ms TTL=125
Reply from 12.0.2.2: bytes=32 time=2ms TTL=125

Ping statistics for 12.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 129.70.1.2

Pinging 129.70.1.2 with 32 bytes of data:

Reply from 129.70.1.2: bytes=32 time=2ms TTL=125
Reply from 129.70.1.2: bytes=32 time=1ms TTL=125
Reply from 129.70.1.2: bytes=32 time=1ms TTL=125
Reply from 129.70.1.2: bytes=32 time=1ms TTL=125

Ping statistics for 129.70.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>
```

```
QEMU (Win-5)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 12.0.0.2

Pinging 12.0.0.2 with 32 bytes of data:

Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125

Ping statistics for 12.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 12.0.1.2

Pinging 12.0.1.2 with 32 bytes of data:

Reply from 12.0.1.2: bytes=32 time=3ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 12.0.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 12.0.2.2

Pinging 12.0.2.2 with 32 bytes of data:

Reply from 12.0.2.2: bytes=32 time=2ms TTL=125
Reply from 12.0.2.2: bytes=32 time=2ms TTL=125
Reply from 12.0.2.2: bytes=32 time=2ms TTL=125
Reply from 12.0.2.2: bytes=32 time=1ms TTL=125

Ping statistics for 12.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 129.70.0.2

Pinging 129.70.0.2 with 32 bytes of data:

Reply from 129.70.0.2: bytes=32 time=7ms TTL=125
Reply from 129.70.0.2: bytes=32 time=2ms TTL=125
Reply from 129.70.0.2: bytes=32 time=3ms TTL=125
Reply from 129.70.0.2: bytes=32 time=1ms TTL=125

Ping statistics for 129.70.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 3ms

C:\Documents and Settings\user>
```